

Aakarshan Khadka

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Profile

Backend Engineer with 1.5+ years of experience building Python-based serverless and cloud-native systems on AWS. Proven ability to design and ship production RESTful APIs — including a multi-tenant SaaS platform, a high-volume property listing backend, and AI-powered services. Hands-on with Lambda, DynamoDB, EC2, RDS, and PostgreSQL. Detail-oriented, comfortable with distributed systems, and eager to grow within a product-minded engineering team.

Experience

AI Engineer

Remote, US

IdeaboxAI (via AZMinds)

Nov 2025 – Present

- Built and maintained Python backend services for Actian's VectorAI DB SDK, implementing gRPC and REST interfaces for high-performance database operations.
- Engineered RESTful APIs enabling AI agents to interact with structured databases and external services via Text-to-SQL and MCP tool integrations.
- Optimized AWS infrastructure costs, reducing monthly cloud spend by 25% through resource rightsizing and configuration improvements.
- Built a custom document parsing engine supporting PDF, DOCX, Markdown, and image formats, optimizing structural data extraction for RAG-based retrieval pipelines.

Backend Developer & DevOps Engineer

Remote

Press1

Jul 2025 – Jan 2026

- Served as founding engineer — architected backend infrastructure from scratch including AWS account setup, system design, and CI/CD pipelines.
- Built and deployed FastAPI backend services for a mobile application and a POS system on EC2 with full Docker containerization.
- Managed domain configuration, SSL, server infrastructure, and deployment pipelines for production environments.

Backend Engineer

Baneshwor, Kathmandu

AZMinds

Dec 2024 – Oct 2025

- Built backend for [Agent Next Door](#) , a Singapore property listing platform handling hundreds of thousands of property and agent records; optimized RESTful APIs and query performance for high-volume data retrieval.
- Integrated Google Maps API, OneMap API, and Singapore government data APIs for property location enrichment and data aggregation.
- Architected SFDS, a multi-tenant SaaS driving school management platform with admin, instructor, and student portals built on Python, FastAPI, and PostgreSQL.
- Integrated Stripe and Authorize.net payment gateways with multi-provider support; implemented Google Calendar OAuth sync for real-time cross-user appointment management.
- Built wildcard domain routing via FastAPI middleware with automatic tenant resolution from database lookups; auto-provisioned 8 fully editable pages per school on onboarding.
- Automated 9+ email notification workflows — appointment confirmations, cancellations, feedback, and certificate generation — with admin-customizable HTML templates.
- Deployed full stack on AWS EC2 via Docker; utilized Lambda, DynamoDB, and RDS with cloud cost optimization.

Backend Intern

Baneshwor, Kathmandu

AZMinds

Sept 2024 – Dec 2024

- Gained hands-on experience with Docker, AWS (Lambda, EC2, RDS, DynamoDB), and cloud-native environ-

ments.

- Collaborated with cross-functional teams on client project delivery, working with real-time data pipelines.

Freelance & Independent Work

Tech Lead

Manaka

Kathmandu, Nepal

Dec 2025 – Present

- Sole developer for a full-stack order management system handling 300+ orders and 250+ customers; built end-to-end from React/TypeScript/Vite frontend to backend API and PostgreSQL database.
- Integrated courier logistics and built an analytics dashboard tracking revenue, expenses, and COGS in real time; reduced order processing time by approximately 80%.

Freelance AI Engineer

Private Client

Remote

Feb 2025

- Built a local email RAG system with hybrid search combining FAISS semantic similarity and TF-IDF keyword scoring at a 0.7/0.3 weighted ratio.
- Implemented cross-encoder reranking (ms-marco-MiniLM-L-6-v2) on top-20 candidates, achieving ~90% average search accuracy.
- Delivered end-to-end solo within 2.5 weeks; live at <https://inboxfind.streamlit.app/> 

Projects

AI Textbook Generation Pipeline

- FastAPI backend orchestrating multiple AI agents (Planner, Writer, Reviewer, Reviser) with a configurable review loop; approved chapters persisted to SQLite.
- Demonstrates async Python, modular agent design, and production API patterns. Live demo available.

RescueSathi — Mobile Application

- AI-powered rescue platform using YOLO for localization and ResNet50 for severity classification; generates geotagged reports for rescue organizations.

E-Commerce Admin Portal

- Full-stack order management and analytics platform built for a live handmade jewelry brand handling 300+ orders and 250+ customers.
- Features a 5-phase fulfillment pipeline, courier logistics integration, real-time COGS and revenue tracking, and product catalog — built solo in React, TypeScript, and Vite.

Education

Islington College, London Metropolitan University

BSc (Hons) Computing with Artificial Intelligence

Sept 2023 – Present

Malpi Institute

A Levels

2022

Skills

Programming Languages: Python, SQL, Bash, Dart, HTML, CSS

Backend: FastAPI, Django REST Framework, SQLAlchemy, RESTful API Design

Databases: PostgreSQL, DynamoDB, RDS, Qdrant

Cloud & DevOps: AWS (Lambda, EC2, S3, DynamoDB, RDS, SNS, API Gateway), Docker, NGINX, Git

AI/ML: LLM, RAG, Agents, MCP, LangChain, PyTorch, Scikit-learn

References

Pradip Bhandari (CTO, AZMinds)